

ME23312 ENGINEERING THERMODYNAMICS

Unit I – Basics, Zeroth and First Law · Presentation 4 of 10

The Zeroth Law and Temperature

Thermal equilibrium · the Zeroth law · thermometers and thermometric properties · the triple point · Kelvin, Celsius, Fahrenheit and Rankine scales

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In this presentation

1. Thermal equilibrium – diathermic and adiabatic walls
 2. The Zeroth law of thermodynamics
 3. Why "zeroth" – and what the law gives us
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1. The constant-volume gas thermometer
 2. The triple point and the Kelvin scale
 3. The four temperature scales and their conversion
 4. Using temperatures correctly

1

Thermal equilibrium and the Zeroth law

Thermal equilibrium

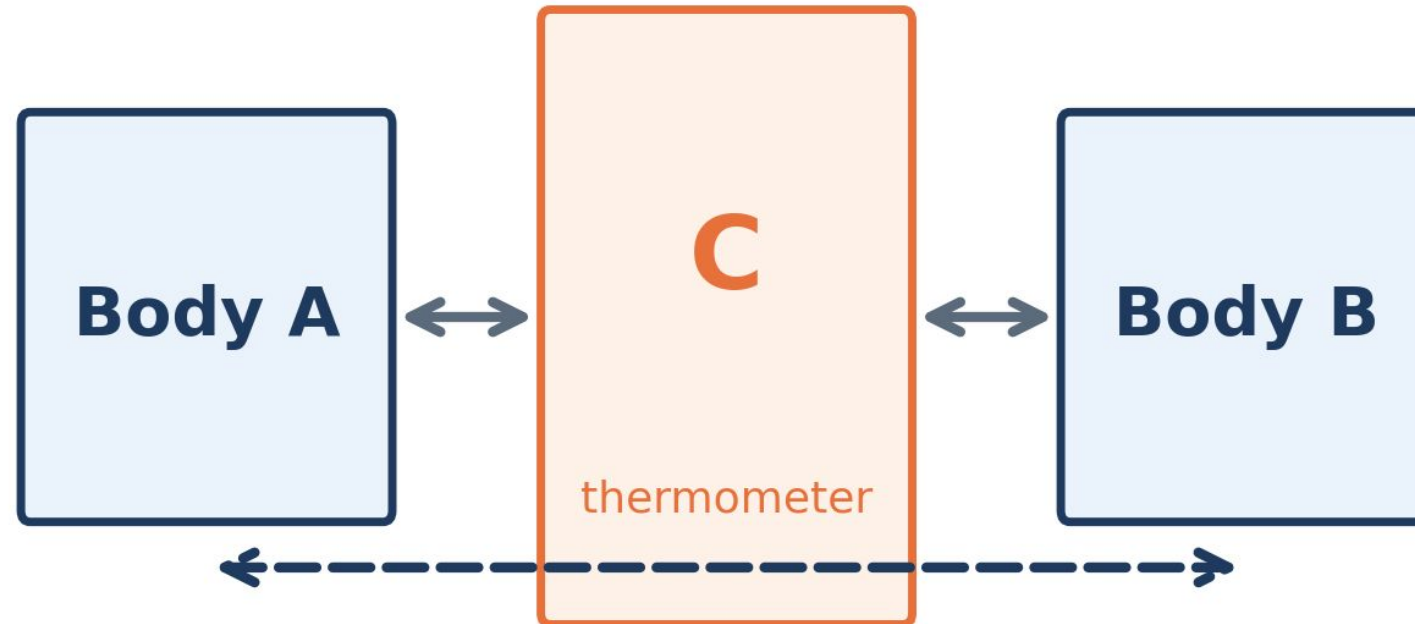
- A hot body and a cold body in contact through a **diathermic** (heat-conducting) wall change until no further change takes place – they are then in **thermal equilibrium**
- Bodies separated by an **adiabatic** wall (perfect insulator) do not interact thermally
- Thermal equilibrium is reached when the temperatures are equal

The Zeroth law of thermodynamics

If two bodies are each in thermal equilibrium with a third body, they are in thermal equilibrium with each other.

- The third body is a **thermometer**; the property they share is **temperature**
- Formulated by R. H. Fowler in 1931 – after the first and second laws, but logically prior to them, hence "zeroth"

The Zeroth law in a picture



A and B are then in thermal equilibrium - they have the same temperature

If A is in thermal equilibrium with C, and B is in thermal equilibrium with C, then A and B are in thermal equilibrium

What the Zeroth law gives us

- If a thermometer reads the same value in contact with A and with B, then A and B are in thermal equilibrium – they have the **same temperature**
- **Temperature** is the property that decides whether two bodies are in thermal equilibrium
- Heat flows from the body at the higher temperature to the body at the lower temperature until the temperatures are equal

KEY POINT The Zeroth law is the basis of all temperature measurement.

2

Thermometers

and thermometric properties

Thermometers and their thermometric properties

Thermometer	Thermometric property
Liquid-in-glass (mercury)	Length of the liquid column
Constant-volume gas thermometer	Pressure of the gas
Resistance thermometer	Electrical resistance of a platinum wire
Thermocouple	Electromotive force (e.m.f.)
Pyrometer	Intensity of radiation

The constant-volume gas thermometer

- The most fundamental thermometer: **all gases at low pressure give the same reading**
- Leads directly to the absolute (Kelvin) temperature scale
- $T = 273.16 \times (p/p_{tp})$, where p_{tp} is the gas pressure at the triple point of water
- Its zero is the **absolute zero** of temperature

3

Temperature scales

The triple point and the Kelvin scale

- Modern scales use a **single fixed point**: the triple point of water – ice, liquid water and water vapour in equilibrium at 0.01 °C and 0.6113 kPa
- Assigned the value **273.16 K** exactly on the Kelvin (absolute, thermodynamic) scale
- Ice point 273.15 K (0 °C); steam point 373.15 K (100 °C) at 1 atm

Fixed points on the four common scales

Scale	Absolute zero	Ice point	Triple point of water	Steam point
Kelvin (K)	0	273.15	273.16	373.15
Celsius (°C)	−273.15	0	0.01	100
Fahrenheit (°F)	−459.67	32	32.02	212
Rankine (°R)	0	491.67	491.69	671.67

Converting between the scales

$$\begin{aligned}
 T \text{ (K)} &= t \text{ (}^\circ\text{C)} + 273.15 & T \text{ (}^\circ\text{R)} &= t \text{ (}^\circ\text{F)} + 459.67 \\
 t \text{ (}^\circ\text{F)} &= 1.8 t \text{ (}^\circ\text{C)} + 32 & T \text{ (}^\circ\text{R)} &= 1.8 T \text{ (K)}
 \end{aligned}$$

- A temperature **difference** of 1 K equals a difference of 1 $^\circ\text{C}$; a difference of 1 $^\circ\text{R}$ equals 1 $^\circ\text{F}$
- Kelvin and Rankine are absolute scales – they start from absolute zero

Using temperatures correctly

- Every thermodynamic relation needs **absolute** temperature: $pV = mRT$, the Carnot efficiency, the definition of entropy
- Always convert Celsius values to kelvin before substituting
- Temperature differences may be left in $^{\circ}\text{C}$, since $\Delta T (\text{K}) = \Delta t (^{\circ}\text{C})$

KEY POINT A gas at 27°C is at 300 K – using 27 in $pV = mRT$ gives an answer that is wrong by a factor of eleven.

Summary

- Thermal equilibrium: no change when bodies are in contact through a diathermic wall
- Zeroth law: two bodies each in thermal equilibrium with a third are in equilibrium with each other
- Temperature is the property shared by bodies in thermal equilibrium; the third body is the thermometer
- Thermometric properties: column length, gas pressure, resistance, e.m.f., radiation
- Triple point of water = 273.16 K exactly; the gas thermometer gives $T = 273.16 \, p/p_{tp}$
- $T \text{ (K)} = t \text{ (}^\circ\text{C)} + 273.15$; $T \text{ (}^\circ\text{R)} = t \text{ (}^\circ\text{F)} + 459.67$; use kelvin in every formula

Next: Energy interactions – work and heat

- The thermodynamic definition of work and the sign convention
- Forms of work: displacement, paddle-wheel, shaft, electrical, flow
- Heat, adiabatic processes and the comparison of heat with work